Ethics & Detection Report — Port Scanning Activity

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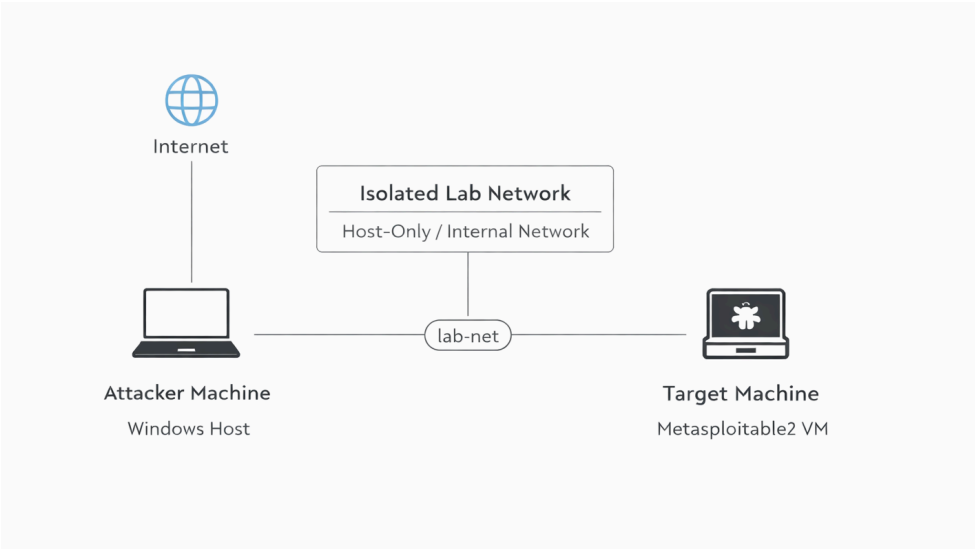
1. Test Environment (Lab-Only Confirmation)

All testing was conducted **exclusively in a controlled laboratory environment** to ensure no unauthorized systems were affected.



Environment setup:

- **Attacker machine:** Windows host
- **Target machine:** Metasploitable2, a intentionally vulnerable VM
- **Virtualization:** VirtualBox (Host-Only / Internal Network)
- **Network isolation:** The lab network was **not connected to the public internet** and only allowed communication between the virtual machines.

This setup ensured:

- No external systems were scanned.
- No production infrastructure was impacted.
- All activity was performed in a **safe cybersecurity training environment**.

Lab-only testing is a **standard ethical practice** in penetration testing training and aligns with the principles of responsible security research.

2. Port Scanning From a Defender's Perspective

From a defender's point of view, a port scan appears as **a series of rapid connection attempts to multiple ports on a target system**. Security analysts can observe these activities using tools like **Wireshark**, intrusion detection systems (IDS), or firewall logs.

No.	Time	Source	Destination	Protocol	Length	Info
382	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	36066 → 29 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
383	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	66	22 → 35486 [ACK] Seq=1 Ack=2 Win=5824 Len=0 TSval=1846413 TSec
384	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	66	23 → 48734 [ACK] Seq=1 Ack=2 Win=5824 Len=0 TSval=1846413 TSec
385	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	66	25 → 45708 [ACK] Seq=1 Ack=2 Win=5824 Len=0 TSval=1846413 TSec
386	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	29 → 36066 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
387	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	45652 → 30 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
388	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	30 → 45652 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
389	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	50966 → 31 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
390	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	31 → 50966 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
391	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	35508 → 32 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
392	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	32 → 35508 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
393	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	35010 → 33 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
394	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	33 → 35010 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
395	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	51960 → 34 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
396	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	34 → 51960 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
397	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	56292 → 35 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
398	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	35 → 56292 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
399	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	44164 → 36 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
400	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	36 → 44164 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
401	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	47004 → 37 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
402	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	37 → 47004 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
403	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	55804 → 38 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
404	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	38 → 55804 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
405	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	40332 → 39 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
406	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	39 → 40332 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
407	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	33692 → 40 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
408	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	40 → 33692 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
409	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	53816 → 41 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
410	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	41 → 53816 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
411	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	38422 → 42 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
412	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	42 → 38422 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
413	2026-03-11 22:19:32…	192.168.1.17	192.168.1.28	TCP	74	38526 → 43 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSva
414	2026-03-11 22:19:32…	192.168.1.28	192.168.1.17	TCP	54	43 → 38526 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

From this screenshot: Many RST responses if ports are closed.

Indicators defenders may observe:

Multiple TCP SYN packets sent to different ports.

- Packets originating from the same source IP.
- Very short time intervals between packets.
- Many RST responses if ports are closed.

This traffic pattern is a strong indicator of **reconnaissance activity**.

Defenders may also notice:

- A single host probing many ports sequentially (1,2,3,....1024)
- Hundreds of ports tested in seconds(1000 ports tested in 2 seconds)
- Repeated attempts across the same network

Security monitoring systems may trigger alerts such as:

- "Possible port scan detected"
- "Reconnaissance activity"
- "Multiple connection attempts to closed ports"

Modern SOC tools **(SIEM, IDS**) often detect these behaviors automatically.

3. Why Authorization Matters Before Running Any Scanner

1. **Legal Responsibility**

   Running a port scanner without permission is **illegal in many jurisdictions** and violates cybersecurity ethics.

   In Belgium:

   The key law is **Article 550bis of the Belgian Criminal Code**, which criminalizes unauthorized access to computer systems.

   Main rule:

   - Anyone who **knowingly accesses a computer system without authorization** can be punished with:
     - 3 months to 1 year imprisonment
     - and/or a fine.

   Additional provisions increase penalties if the attacker:

   - accesses or copies data
   - causes damage to the system
   - acts with intent to defraud or harm.

   Port scanning can be interpreted as **unauthorized reconnaissance**.

   Even if no exploitation occurs, **the scan itself can trigger legal consequences**.

2. **Impact on Production Systems**

   Scanning can:

   - Trigger security alarms
   - Overload fragile services
   - Crash poorly configured systems
   - Disrupt production environments

   Organizations therefore restrict scanning to **authorized security assessments**.

3. **Ethical Penetration Testing Standards**

   Professional penetration testers always operate under **explicit written permission**, usually through:

   - Rules of Engagement (RoE)
   - Scope documentation
   - Authorization letters

   These documents define:

- Which systems can be tested
- What techniques are allowed
- When testing can occur

Without this authorization, a scan may be treated as **hostile activity**.

4. Conclusion

This exercise demonstrated how port scanning functions and how it appears from a defensive monitoring perspective. Conducting the test in an isolated lab environment ensured that the activity remained **safe, legal, and educational**.

Understanding both sides of the process — attacker reconnaissance and defender detection — is essential for cybersecurity professionals, particularly in **incident response and threat detection roles**.